

## MANTA, Ecuador

WMO: 841170

Lat: 0.950S

Lon: 80.683W

Elev: 48

StdP: 14.67

Time Zone: -5.00 (W05)

Period: 95-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 10            | 67.7       | 68.1 | 62.2                          | 83.9 | 72.1 | 62.6 | 85.1 | 71.9 | 20.1                  | 78.0 | 18.6 | 77.6 | 6.9                   | 210  | 0.320 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 3             | 9.3                    | 87.7            | 76.3 | 86.2 | 76.3 | 85.7 | 76.2 | 79.4                | 84.1 | 78.8 | 83.5 | 78.1 | 82.9 | 10.3                 | 210  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 77.5                            | 143.3 | 82.2 | 77.2 | 141.7 | 81.9 | 76.7 | 139.5 | 81.5 | 43.0          | 85.0 | 42.3 | 83.6 | 41.6 | 83.1 | 84.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 17.8              | 16.2 | 14.6 | 66.4                       | 91.4 | 1.8                | 1.4 | 65.1                                               | 92.5 | 64.0       | 93.3 | 63.0       | 94.1 | 61.6       | 95.1 |    |    |
|                   |      |      | 63.3                       | 80.4 | 2.7                | 2.0 | 61.3                                               | 81.9 | 59.7       | 83.0 | 58.2       | 84.2 | 56.2       | 85.7 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 77.5  | 79.0  | 79.0  | 79.4  | 79.0  | 78.8  | 77.2  | 76.4  | 75.9  | 75.9  | 75.7  | 76.3  | 77.8 | (6)  |
| (7)  |                                                                   | DBStd       | 2.59  | 2.07  | 1.63  | 1.53  | 2.00  | 2.22  | 2.57  | 2.57  | 2.36  | 2.37  | 2.08  | 2.27  | 2.27 | (7)  |
| (8)  |                                                                   | HDD50       | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (8)  |
| (9)  |                                                                   | HDD65       | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (9)  |
| (10) |                                                                   | CDD50       | 10046 | 898   | 813   | 911   | 869   | 894   | 816   | 818   | 802   | 776   | 796   | 790   | 863  | (10) |
| (11) |                                                                   | CDD65       | 4571  | 433   | 393   | 446   | 419   | 429   | 366   | 353   | 337   | 326   | 331   | 340   | 398  | (11) |
| (12) |                                                                   | CDH74       | 27790 | 3282  | 3138  | 3728  | 3320  | 3138  | 2008  | 1615  | 1380  | 1265  | 1189  | 1431  | 2296 | (12) |
| (13) |                                                                   | CDH80       | 4793  | 552   | 532   | 717   | 641   | 660   | 335   | 249   | 200   | 177   | 142   | 212   | 376  | (13) |
| (14) | Wind                                                              | WSAvg       | 7.6   | 6.6   | 5.1   | 5.2   | 5.7   | 7.0   | 8.0   | 8.5   | 8.7   | 9.0   | 9.1   | 9.2   | 8.6  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 19.1  | 3.1   | 5.1   | 4.6   | 2.7   | 0.9   | 0.3   | 0.1   | 0.1   | 0.1   | 0.1   | 0.4   | 0.7  | (15) |
| (16) |                                                                   | PrecMax     | 65.8  | 14.7  | 12.4  | 14.0  | 9.0   | 11.0  | 4.3   | 0.9   | 1.6   | 1.3   | 1.4   | 7.7   | 10.5 | (16) |
| (17) |                                                                   | PrecMin     | 5.4   | 0.4   | 1.9   | 0.3   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0  | (17) |
| (18) |                                                                   | PrecStd     | 14.5  | 3.2   | 3.0   | 3.7   | 2.8   | 2.4   | 0.9   | 0.2   | 0.3   | 0.2   | 0.3   | 1.5   | 2.1  | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 88.1  | 86.4  | 87.5  | 88.1  | 88.1  | 87.7  | 86.4  | 86.4  | 87.2  | 86.1  | 86.3  | 88.2 | (19) |
| (20) |                                                                   | MCWB        | 75.0  | 77.5  | 78.5  | 77.8  | 76.6  | 76.3  | 75.3  | 74.6  | 74.9  | 74.6  | 74.3  | 75.1  | (20) |      |
| (21) |                                                                   | 2%          | DB    | 85.9  | 85.7  | 86.0  | 86.1  | 86.2  | 85.6  | 84.4  | 84.2  | 84.1  | 83.0  | 84.3  | 85.8 | (21) |
| (22) |                                                                   | MCWB        | 75.6  | 77.3  | 77.8  | 77.0  | 76.6  | 76.1  | 74.7  | 74.1  | 74.3  | 73.7  | 74.0  | 75.1  | (22) |      |
| (23) |                                                                   | 5%          | DB    | 84.3  | 84.3  | 84.5  | 84.5  | 84.6  | 83.8  | 82.6  | 82.2  | 82.1  | 81.0  | 82.5  | 83.9 | (23) |
| (24) |                                                                   | MCWB        | 75.6  | 76.7  | 77.1  | 76.4  | 76.1  | 75.3  | 74.2  | 73.4  | 73.4  | 73.0  | 73.6  | 74.4  | (24) |      |
| (25) |                                                                   | 10%         | DB    | 82.7  | 82.8  | 83.9  | 83.5  | 83.9  | 82.1  | 80.7  | 80.4  | 80.2  | 79.3  | 80.7  | 82.2 | (25) |
| (26) |                                                                   | MCWB        | 74.9  | 76.1  | 76.9  | 76.0  | 75.8  | 74.4  | 73.3  | 72.7  | 72.8  | 72.3  | 72.9  | 73.7  | (26) |      |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 79.2  | 80.1  | 80.6  | 79.6  | 79.9  | 78.8  | 77.4  | 76.4  | 76.8  | 76.5  | 77.9  | 78.8 | (27) |
| (28) |                                                                   | MDCB        | 84.1  | 84.2  | 84.9  | 85.0  | 85.4  | 84.8  | 83.8  | 83.5  | 83.8  | 83.2  | 82.9  | 84.9  | (28) |      |
| (29) |                                                                   | 2%          | WB    | 78.2  | 79.0  | 79.4  | 78.5  | 78.5  | 77.2  | 75.9  | 74.8  | 75.3  | 74.9  | 76.2  | 77.1 | (29) |
| (30) |                                                                   | MDCB        | 82.9  | 83.1  | 83.6  | 83.7  | 84.2  | 83.1  | 82.2  | 82.1  | 81.7  | 81.4  | 81.4  | 82.3  | (30) |      |
| (31) |                                                                   | 5%          | WB    | 77.1  | 78.2  | 78.6  | 77.6  | 77.4  | 76.0  | 74.7  | 73.9  | 74.1  | 73.9  | 74.8  | 75.9 | (31) |
| (32) |                                                                   | MDCB        | 81.8  | 82.3  | 82.8  | 82.8  | 83.0  | 81.6  | 80.6  | 80.3  | 80.1  | 79.9  | 80.2  | 81.3  | (32) |      |
| (33) |                                                                   | 10%         | WB    | 76.2  | 77.3  | 77.9  | 76.8  | 76.4  | 74.8  | 73.9  | 72.9  | 72.9  | 72.8  | 73.6  | 74.6 | (33) |
| (34) |                                                                   | MDCB        | 80.9  | 81.4  | 82.0  | 81.8  | 81.8  | 80.1  | 79.5  | 78.6  | 78.5  | 78.1  | 79.1  | 80.2  | (34) |      |
| (35) | Mean Daily Temperature Range                                      | MDBR        | 9.1   | 8.8   | 9.3   | 9.7   | 10.1  | 9.9   | 10.0  | 10.4  | 10.1  | 9.8   | 10.5  | 9.9   | (35) |      |
| (36) |                                                                   | 5% DB       | MCDBR | 10.6  | 10.1  | 10.5  | 10.9  | 11.5  | 11.8  | 11.9  | 12.4  | 12.4  | 11.7  | 12.6  | 12.1 | (36) |
| (37) |                                                                   | MCWBR       | 4.9   | 5.0   | 5.3   | 5.3   | 5.4   | 5.6   | 5.5   | 5.6   | 5.6   | 5.7   | 5.8   | 5.4   | (37) |      |
| (38) |                                                                   | 5% WB       | MCDBR | 9.7   | 9.4   | 9.7   | 10.2  | 10.8  | 11.0  | 10.9  | 11.4  | 11.2  | 11.1  | 11.5  | 10.9 | (38) |
| (39) |                                                                   | MCWBR       | 5.2   | 5.5   | 5.6   | 5.5   | 5.8   | 5.8   | 5.7   | 5.8   | 5.9   | 5.9   | 6.2   | 5.8   | (39) |      |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.439 | 0.459 | 0.467 | 0.479 | 0.480 | 0.471 | 0.473 | 0.499 | 0.525 | 0.485 | 0.484 | 0.474 | (40) |      |
| (41) |                                                                   | taud        | 2.396 | 2.334 | 2.311 | 2.271 | 2.273 | 2.303 | 2.304 | 2.210 | 2.120 | 2.248 | 2.241 | 2.279 | (41) |      |
| (42) |                                                                   | Ebn at Noon | 284   | 279   | 274   | 264   | 256   | 254   | 256   | 255   | 255   | 268   | 268   | 272   | (42) |      |
| (43) |                                                                   | Edh at Noon | 40    | 43    | 43    | 44    | 42    | 40    | 41    | 46    | 52    | 46    | 46    | 44    | (43) |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 1426  | 1475  | 1616  | 1643  | 1419  | 1165  | 1114  | 1172  | 1229  | 1235  | 1320  | 1392  | (44) |      |
| (45) |                                                                   | RadStd      | 126   | 114   | 119   | 116   | 118   | 142   | 102   | 104   | 116   | 118   | 115   | 115   | (45) |      |

## Historical Trends

|      |                       | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-----------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                       |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only          | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (47) | Regional (1 neighbor) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page